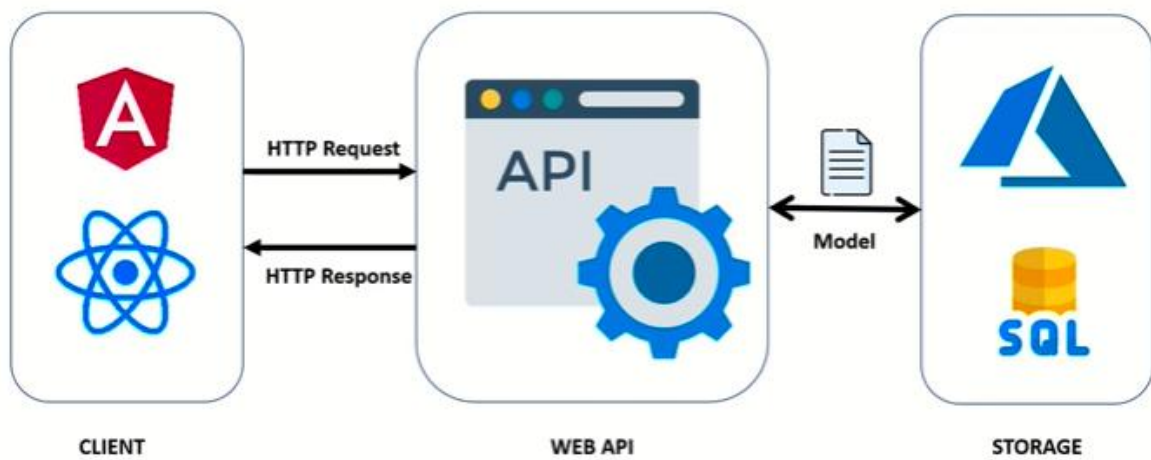


Web API Architecture



SDK

An SDK (Software Development Kit) includes everything you need to build and run .NET core apps using command-line tools and editor (Visual Studio)

.NET Core CLI (command Linet Interface)

A cross-platform toolchain for developing, building, running and publishing .Net applications.

CLASS AND OBJECT

1. Class is the main part of OOps programming, it's a blueprint of an object.
2. Classes contain the variable ,methods to perform the operations.
3. The default accessibility of a class is internal.
4. Private is the default accessibility of class members.

OBJECT IN C#

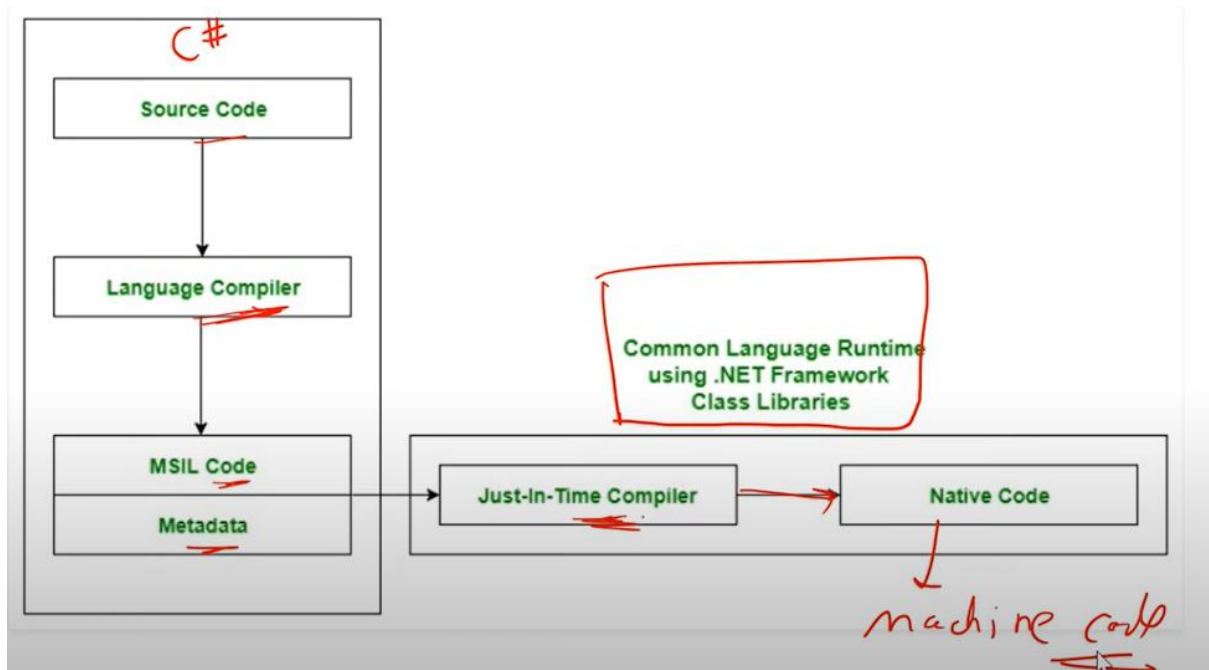
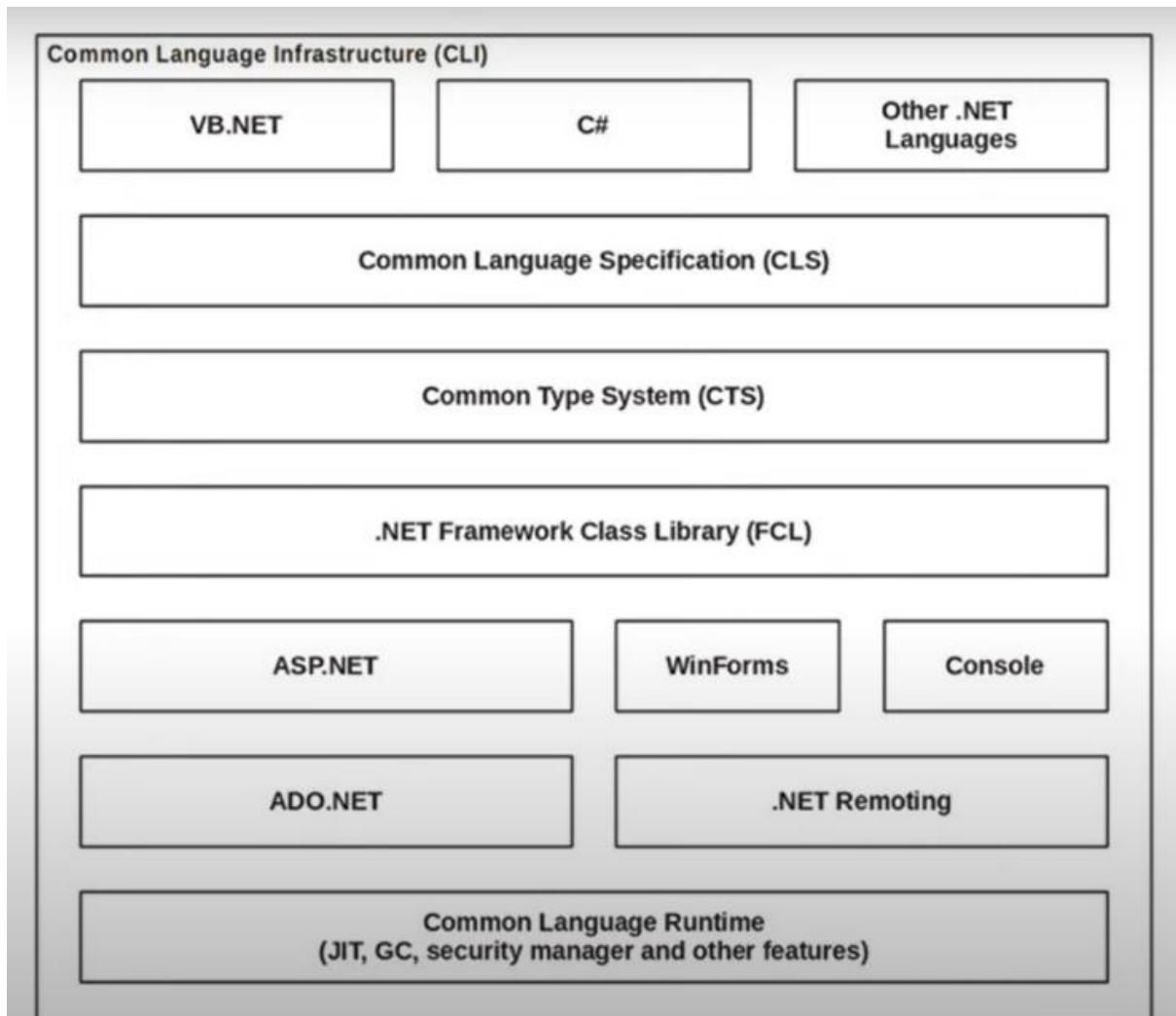
- An object is a combination of related variable and methods.
- An object is an instance of a class"
- A class will not occupy any memory space, hen an object is created without the new operator, memory will not be allocated in the heap,

ABSTRACTION

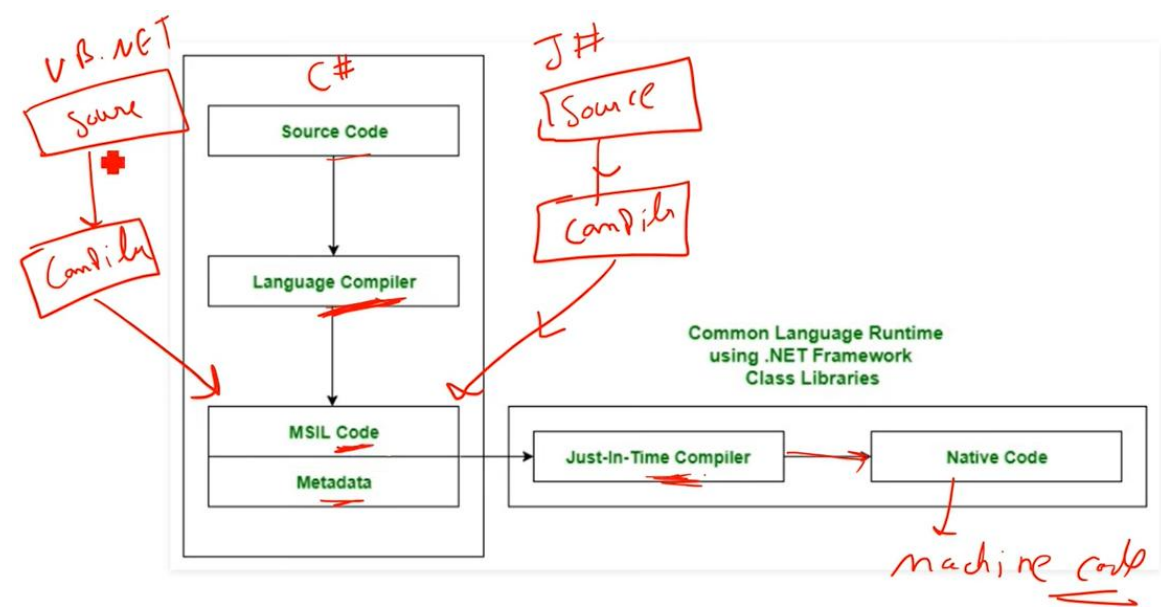
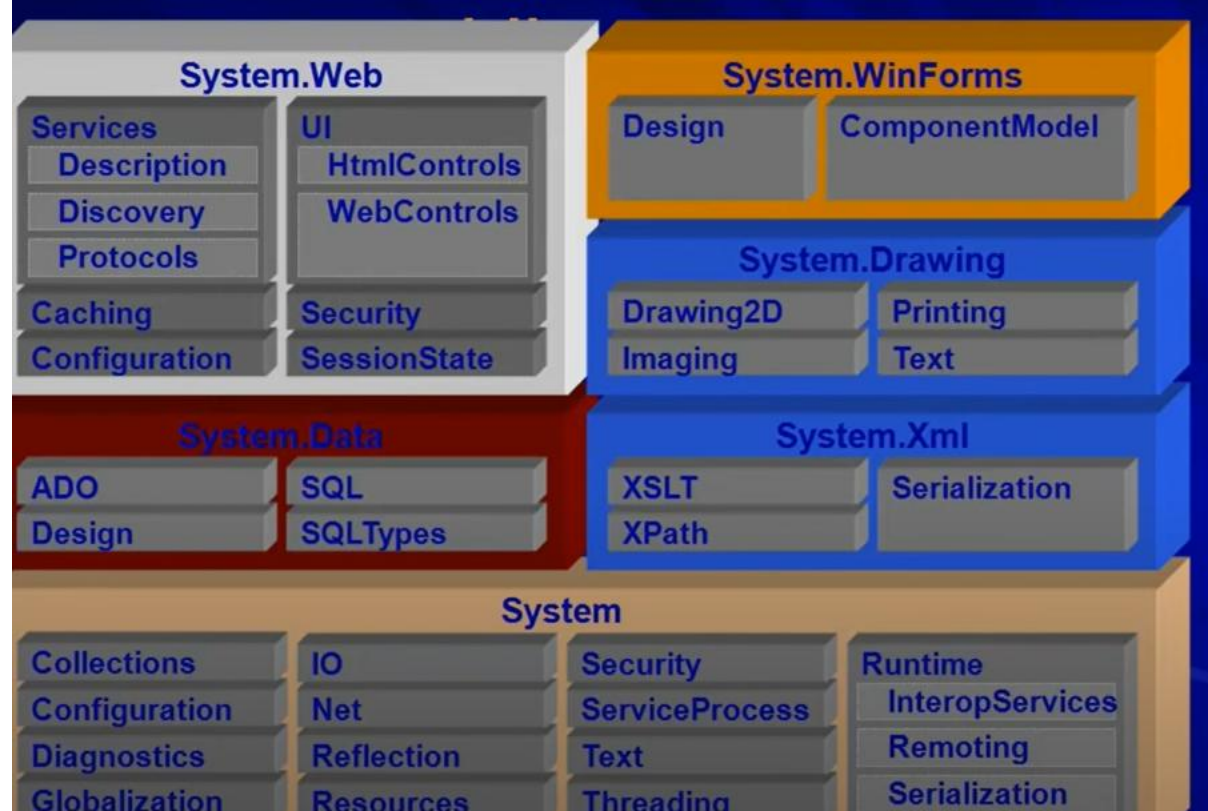
- Abstraction is the act of representing essential information without including background details and explanations.
- Abstraction means putting all the variables and methods in a class that are necessary.

.NET

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T - An overview of .NET Framework Class Library (FCL / BCL) Base class Lib



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BCL

- Contains a set of pre-defined classes that can be used in all types of .net applications & languages for general I/O operations, type conversion, creation of threads etc.
- Ex:
 - Console
 - String
 - StringBuilder
 - Convert
 - Thread
 - Task
 - etc.

ADO.NET

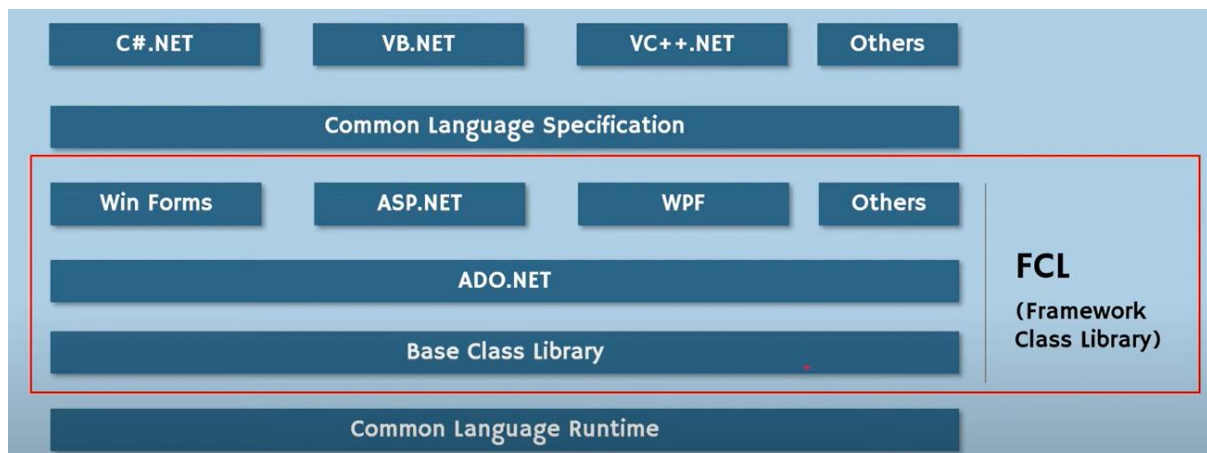
- Contains a set of pre-defined classes that can be used in all types of .net applications & languages for connecting to databases, retrieving data from databases, inserting, updating, deleting rows etc.
- Ex:
 - SqlConnection
 - SqlCommand
 - SqlDataAdapter
 - SqlDataReader
 - etc.

Win Forms

- Contains a set of pre-defined classes that can be used in Windows GUI applications for development of GUI elements such as form, textbox, button, checkbox, radio button, dropdownlist etc.
- Ex:
 - Form
 - Label
 - Button
 - TextBox
 - etc.

ASP.NET

- Contains a set of pre-defined classes that can be used in Web Applications for development of GUI elements such as page, textbox, button, checkbox, radio button, dropdownlist etc.
- Ex:
 - Page
 - Label
 - Button
 - TextBox
- etc.



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CLS

- Contains a set of rules (concepts) that are common to all .net languages such as C#.NET, VB.NET etc.
 - Common rules of CLS:
 - CTS (Common Type System): Contains data types such as Int32, Int64, Single, Double etc.
 - Classes & Objects
 - Reference Variables
 - Method Parameters
 - Generics
- etc.

.Net Framework History

https://www.google.com/search?q=difference+between+.NET+and+.NET+Framework+in+hindi&sca_esv=3a183fcd788f8a90&rlz=1C1GCEU_enIN1073IN1073&udm=7&biw=1366&bih=633&ei=rSB0Z_qPFd6e4-EPwZbnuQo&ved=0ahUKEwi6mvS8u9KKAxVezzGHHUHOacQ4dUDCBE&uact=5&oq=difference+between+.NET+and+.NET+Framework+in+hindi&gs_lp=EhZnd3Mtd2l6LW1vZGVsZXNzLXZpZGVvIjNkaWZmZXJlbnNlIGJldHdlZW4gLk5FVCBhbmQgLk5FVCBGcmFtZXdvcmmsgaW4gaGluZGkyBRAhGKABSPkSUIsHWPYQcAF4AJABAjgBswGgAZ4JqgEDMC45uAEDyAEA-AEBmAIKoAK9CcICCBAAAGAcYChgewglGEAAYCBgewglLEAAYgAQYhgMYigXCAggQABiABBiiBMICBRAhGJ8FwglEECEYFcICBxAhGKABGAqYAwCIBgGSBwMxLjmgB-Yp&sclient=gws-wiz-modeless-video#fpstate=ive&vld=cid:5f65c9e0,vid:WnlMdf6T-iY,st:0

.NET Framework

- Drawback: Only ran on windows.
- Last version is 4.8
- Still supported from Microsoft.
- Still in use in many code bases.

The [Mono Project \(mono/mono\)](#) ('original mono') has been an important part of the .NET ecosystem since it was launched in 2001. Microsoft became the steward of the Mono Project when it acquired Xamarin in 2016.

The last major release of the Mono Project was in July 2019, with minor patch releases since that time. The last patch release was February 2024.

Cross platform run/port from .NET to Linux

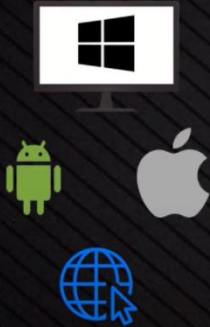
All library of .net framework could not be move to Mono because of .net framework library have dependencies on Windows platforms.

The Mono Project

- Common Language Infrastructure (CLI)
- Open standards: ECMA-335 and ECMA-334
- Mono: A project to port .NET to Linux
- Production ready but limited

2013 and onward the below: This is due to run the project on every platform like Android, Linux, Apple, Windows

.NET Everywhere

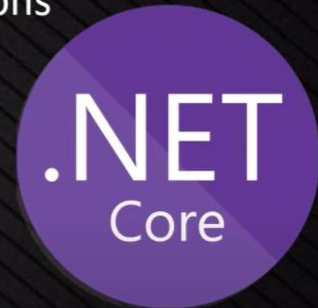


- .NET Framework
- .NET Micro Framework
- Xamarin
- Silverlight

2014 developed:

The Rise of .NET Core

- Complete redesign of the .NET specifications with cross-platform support in mind
- Started in 2014
- First release in 2016
- Ran on Windows, Linux, and macOS



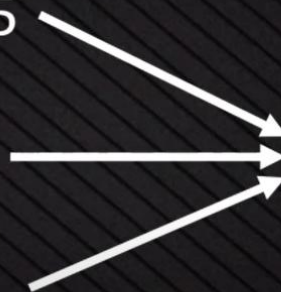
The Need of .NET Standard

.NET Framework 4.5

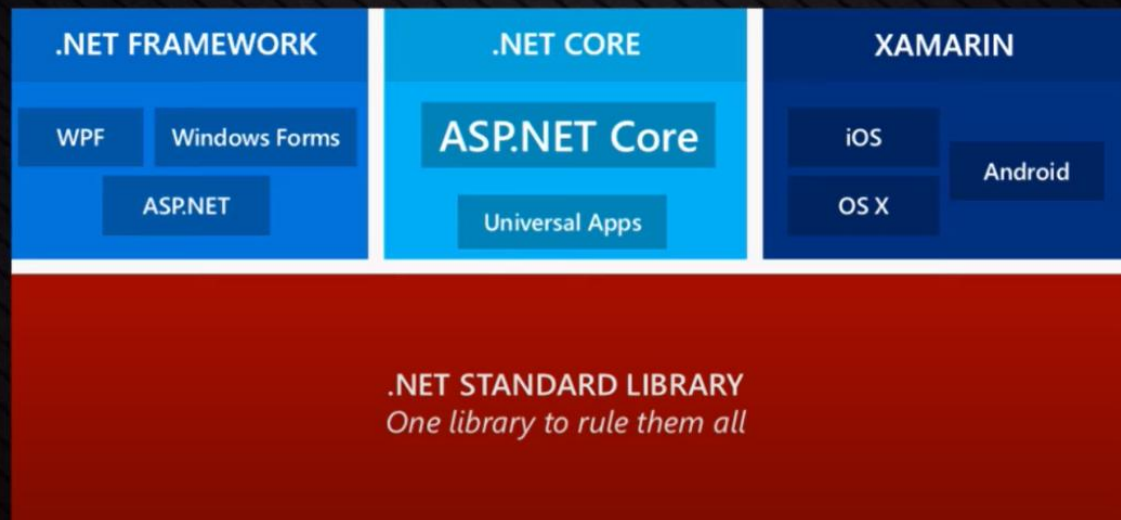
.NET Core 2.1

Mono 4.6

.NET
Standard 1.1



The Need of .NET Standard



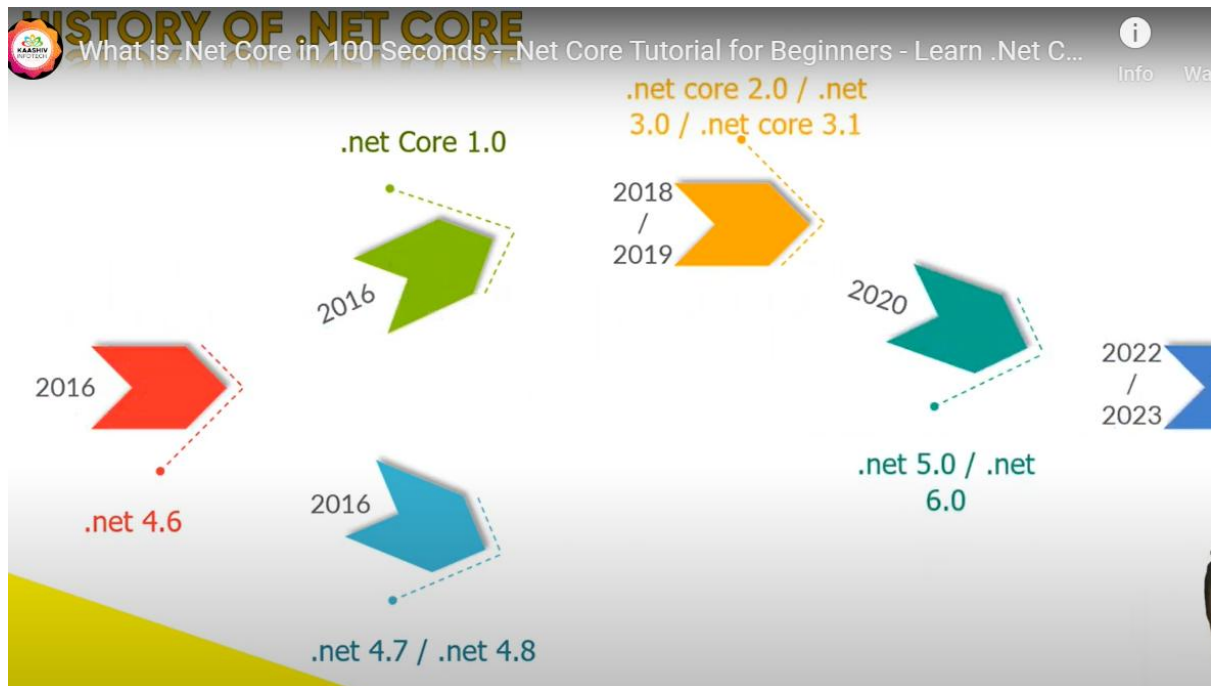
Nov 2022 launch the .NET which replace all the above.

.NET Now

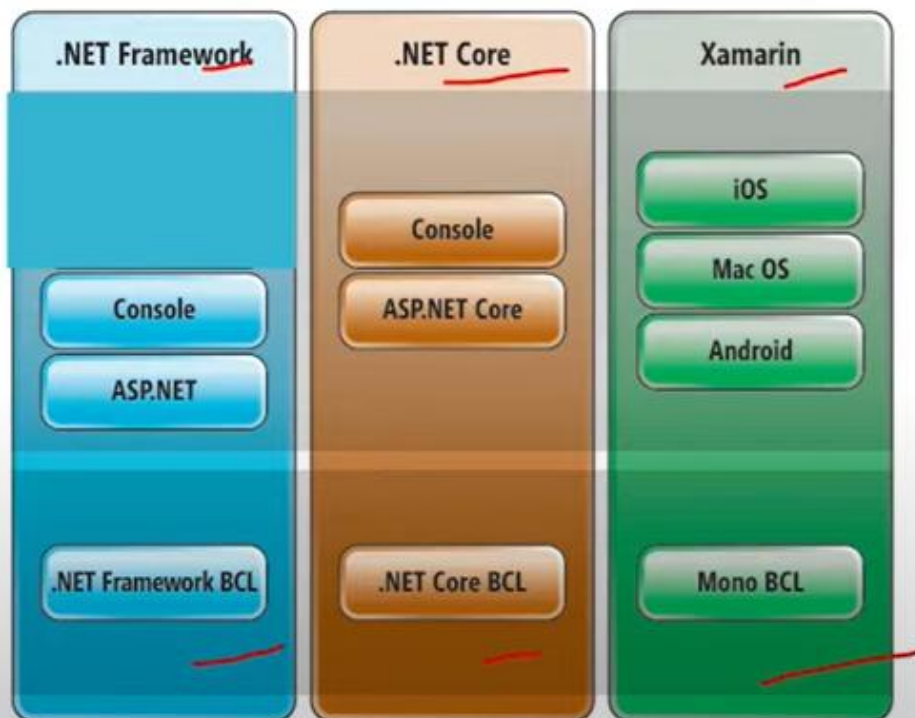
.NET – A unified platform



.NET Core History



- ❖ .NET Core is completely a NEW framework, which is a FREE and OPEN-SOURCE platform developed and maintained by Microsoft.



Introduction

- NET Core is a **free open source** and **cross platform framework**, created for building **modern Cloud based Applications** and every .NET developer feels proud of it.
- Now, there are no boundaries for the **platform**. Now, every .NET developer can say, yes I am **platform independent**, I am using an **open source**.

Introduction

- In the year **2016**, Microsoft has come up with a new revolution, **Microsoft .NET Core 1.0**. People always love ASP.NET, because it's working over **WorldWideWeb**.
- ASP.NET Core is a new open source, cross platform framework to create modern Web based Cloud based systems, which means, now you are not only working for Windows, you can run in Linux, Mac; i.e., wherever you want.

ASP.NET Core features

- Cross platform, open source now runs your app over Linux, Windows, Mac; i.e., wherever you want.
- **Fast Development** - fast work over the Browsers.
- **Work in your editors** - now you can work not only in Visual Studio. You can also choose Visual Studio code.

ASP.NET Core features

- It is based on **Model-View-Controller (MVC)** architecture.

What Is ASP.NET Core?



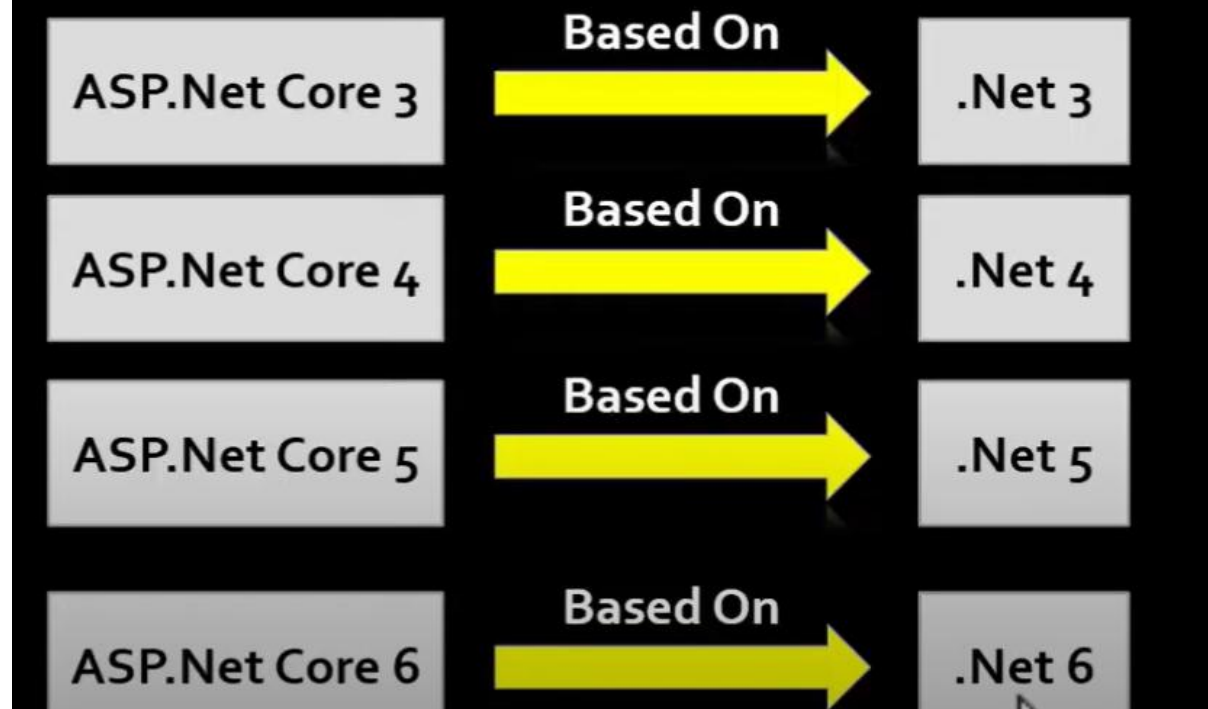
- ASP.NET Core is a new version of ASP.NET, developed by **Microsoft**.
- It is an **open-source framework** for developing web applications and it can be run on **Windows, Mac, or Linux**.
- It has been completely **rewritten from scratch** and it was initially launched as **ASP.NET 5** but then was renamed to **ASP.NET CORE 1.0**.

Introduction



- **Before ASP.NET Core**, Microsoft already had 2 other similar frameworks, these were – **ASP.NET Web Forms (popularly known as just ASP.NET)** and **ASP.NET MVC**.
- The **latest version** of ASP.Net Core is **ASP .Net Core 7.0**. It was released on **February 24, 2023**.

Introduction



Introduction

- From .NET 5.0, the word "Core" is dropped from it's name.
- So all frameworks from now on will be named as .NET 6, .NET 7 and so on.

Pre-Requisite

- HTML
- CSS
- Javascript
- Bootstrap (Optional)
- C# Programming language.
- MVC
- Visual Studio 2017 or higher version
- We use Visual Studio 2022 (Community Version) in this tutorial series.